

# Taeyun Roh

Seoul, Republic of Korea | [txnrd.github.io](https://txnrd.github.io) | [github.com/txnrd](https://github.com/txnrd)

LLM researcher developing AI agents that better perceive their environments through memory systems and visual reasoning.

## Education

**Korea University**  
*M.S. in Computer Science and Engineering*

Mar. 2026 – Present  
Seoul, Republic of Korea

- Master’s student in Computer Science and Engineering.

**Korea University**  
*B.S. in Computer Science and Engineering (GPA: 4.08 / 4.5)*

Mar. 2022 – Feb. 2026  
Seoul, Republic of Korea

- Undergraduate studies focused on systems, machine learning, and software engineering.

## Research Experience

**Data Mining & Information Systems Laboratory, Korea University**  
*Undergraduate Researcher, advised by Prof. Jae Woo Kang*

Mar. 2025 – Present  
Seoul, Republic of Korea

- Research on agentic memory systems for AI agents.

**Network and Mobile System Laboratory, KAIST**  
*Research Intern, advised by Prof. Sung-Ju Lee*

Jan. 2025 – Mar. 2025  
Daejeon, Republic of Korea

- Worked on efficient LLM inference, including on-device LLM inference.

**System Software Laboratory, Korea University**  
*Undergraduate Researcher, advised by Prof. Gyeongsik Yang*

Jan. 2024 – Dec. 2024  
Seoul, Republic of Korea

- Researched federated learning for healthcare applications under resource heterogeneity.

## Selected Publications

**CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents**  
Taeyun Roh, Wonjune Jang, Junha Jung, Jaewoo Kang

ACL 2026 Findings

**Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning**  
Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, Taeyun Roh, Mujeen Sung, Jaewoo Kang

Under Review

**When Choices Become Priors: Contrastive Decoding for Scientific Figure Multiple-Choice QA**  
Taeyun Roh, Eun-yeong Jo, Wonjune Jang, Jaewoo Kang

Under Review

## Industry & Development Experience

**Aigensciences**  
*AI Researcher*

Mar. 2025 – Aug. 2025  
Seoul, Republic of Korea

- Built an LLM-based pipeline for automated hypothesis generation in drug discovery.
- Developed a pediatric counseling chatbot designed for KakaoTalk integration.

**Blurting**  
*Frontend Developer, Co-founder*

Jun. 2023 – Jan. 2025  
Seoul, Republic of Korea

- Co-founded a four-member startup and led end-to-end frontend development.

- Developed and operated a blind-dating application that reached #73 in the Korea App Store Social Networking category.

## Hiconsy

Data Analyst

Jan. 2022 – Jan. 2023  
Seoul, Republic of Korea

- Built mathematical models for optimizing teaching assistant compensation.
- Applied regression analysis to study the relationship between student study time and academic performance.

## Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper “*Analysis on Re-search for Data and Resource Heterogeneity in Split-Federated Learning.*”
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

## Teaching Experience

### Korea University

Teaching Assistant, Machine Learning (COSE 362), instructed by Prof. Jae Woo Kang

Fall 2025  
Seoul, Republic of Korea

## Academic Service

### Conference Reviewer

ARR JAN 2026 (Secondary)

### Workshop Reviewer

EIML@ICML 2026

## Selected Projects & Skills

### Projects

- **Xalute** — Wearable health monitoring app developed with Korea University Anam Hospital for real-time health tracking and alerts.
- **Kakao Pediatric Response Chatbot** — LLM-based pediatric Q&A chatbot developed with Kakao Healthcare and Upstage.

### Skills

- **Research:** Memory system for agents, Multimodal reasoning, LLM inference
- **Development:** Frontend development, applied LLM products, data analysis
- **Language:** English (TOEIC 940)